

17.2.1 (Shankar)

We are given $H = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 x^2 + \lambda x^4$. In terms of the creation and annihilation operators

$$x = \left(\frac{\hbar}{2m\omega}\right)^{1/2} (a + a^\dagger)$$

$$\therefore H' = \lambda \left(\frac{\hbar}{2m\omega}\right)^2 (a + a^\dagger)^4$$

The first order energy shift is

$$E_n' = \langle n | H' | n \rangle$$

$$= \frac{\lambda \hbar^2}{4m^2\omega^2} \left\{ \langle n | (a + a^\dagger)^4 | n \rangle \right\}$$

$$(a + a^\dagger)^4 = a^4 + (a^3 a^\dagger + 3 \text{ other terms})$$

$$+ \left\{ (a^2 a^{\dagger 2} + a a^\dagger a a^\dagger + a a^\dagger a^\dagger a + a^\dagger a a a^\dagger + a^\dagger a a^\dagger a + a^{\dagger 2} a^2) \right\}$$

$$+ (a a^{\dagger 3} + 3 \text{ other terms}) + (a^\dagger)^4$$

Then terms in the curly bracket only contribute to $\langle n | \dots | n \rangle$

and we get

$$E_n' = \frac{\lambda \hbar^2}{4m^2\omega^2} \left\{ \underbrace{(n+2)(n+1)}_{(n^2+3n+2)} + \underbrace{(n+1)^2}_{(n^2+2n+1)} + \underbrace{2n(n+1)}_{(2n^2+2n)} + \underbrace{n^2}_{n^2} + \underbrace{n(n-1)}_{(n^2-n)} \right\}$$

where we used

$$\langle n | a^2 a^{\dagger 2} | n \rangle = (n+1)(n+2)$$

$$\langle n | a a^\dagger a a^\dagger | n \rangle = (n+1)^2$$

$$\langle n | a a^\dagger a^\dagger a | n \rangle = n(n+1)$$

$$\langle n | a^\dagger a a a^\dagger | n \rangle = n(n+1)$$

$$\langle n | a^\dagger a a^\dagger a | n \rangle = n^2$$

$$\langle n | a^{\dagger 2} a^2 | n \rangle = n(n+1)$$

$$E_n' = \frac{3\lambda \hbar^2}{4m^2\omega^2} (1 + 2n + 2n^2)$$

(2) No matter how small λ is we see that -

(2)

$$E_n' \sim \underbrace{(E_{n+1}^0 - E_n^0)}_{\hbar\omega} \quad \text{for large enough } n.$$

Note for large n , $E_n' \sim \frac{3\lambda n^2 \hbar^2}{2m^2 \omega^2}$ ($\lambda n^2 \sim 1$) is possible!

Physically for x very large λx^4 is comparable to or dominates and the potential is no longer like a harmonic oscillator! Thus particles with large energy will experience a different potential.

17.2.2 (Shankar).

The Hamiltonian

$$H = -\gamma \vec{S} \cdot \vec{B}$$

$$= -\frac{\gamma \hbar}{2} B_0 \sigma_3 - \frac{\gamma \hbar}{2} B \sigma_1$$

$$= -\frac{\gamma \hbar}{2} \begin{pmatrix} B_0 & B \\ B & -B_0 \end{pmatrix}$$

Exact Energy

$$-\frac{\gamma \hbar}{2} (B_0^2 + B^2)^{1/2}$$

$$\frac{\gamma \hbar}{2} (B_0^2 + B^2)^{1/2}$$

Exact Eigen function

$$\left(\frac{(B_0 + \sqrt{B_0^2 + B^2})^{1/2}}{2\sqrt{B_0^2 + B^2}} \right) \begin{bmatrix} 1 \\ B/(B_0 + \sqrt{B_0^2 + B^2}) \end{bmatrix}$$

$$\left(\frac{B_0 + \sqrt{B_0^2 + B^2}}{2\sqrt{B_0^2 + B^2}} \right)^{1/2} \begin{bmatrix} B/(B_0 + \sqrt{B_0^2 + B^2}) \\ 1 \end{bmatrix}$$

Expanding the exact energy to order B and B^2 and
eigenvector to order B we see that -

(3)

Energy

$$-\frac{\gamma \hbar}{2} B_0 - \frac{\gamma \hbar B^2}{4 B_0}$$

$$+\frac{\gamma \hbar}{2} + \frac{\gamma \hbar}{4} \frac{B^2}{B_0}$$

Eigen Vector

$$\begin{bmatrix} 1 \\ B \end{bmatrix}$$

$$\begin{bmatrix} B \\ 1 \end{bmatrix}$$

It is easy to verify that the first order
correction to the energy is zero and the
second order energy correction agrees with the
above result. Similarly the first order correction
to the level also agrees with the above result.

17.2.4 (Shankar) . Thomas - Reiche - Kuhn Sum rule .

Given $H = \frac{P^2}{2m} + V(x)$, let us show that -

$$\sum_{n'} (E_{n'} - E_n) |\langle n' | x | n \rangle|^2 = \sum \frac{\hbar^2}{2m}$$

L.H.S is given by

$$\begin{aligned} & \sum_{n'} (E_{n'} - E_n) \langle n | x | n' \rangle \langle n' | x | n \rangle \\ &= \sum_{n'} \langle n | (x H - H x) | n' \rangle \langle n' | x | n \rangle \end{aligned}$$

$$= \sum_{n'} \langle n | [X, H] | n' \rangle \langle n' | X | n \rangle = \langle n | [X, H] X | n \rangle \quad (4)$$

We can also see that the L.H.S is

$$\sum_{n'} \langle n | X | n' \rangle \langle n' | (HX - XH) | n \rangle = - \langle n | X [X, H] | n \rangle$$

$$\begin{aligned} \therefore \text{L.H.S} &= \frac{1}{2} \left\{ \langle n | [X, H] X | n \rangle - \langle n | X [X, H] | n \rangle \right\} \\ &= \frac{1}{2} \langle n | [[X, H], X] | n \rangle \end{aligned}$$

$$\text{Now } [X, H] = i\hbar \frac{P}{m} \quad [[X, H], X] = \frac{i\hbar}{m} [P, X] = \frac{\hbar^2}{m}$$

$$\therefore \text{L.H.S} = \frac{\hbar^2}{2m} = \text{R.H.S} \quad !$$

Note: $[X, P^2] = [X, P]P + P[X, P] = 2i\hbar P$

$$[X, V(X)] = 0$$

$$\text{Hence } [X, H] = \frac{i\hbar P}{m}$$

17.2.3 If proton is a charge distributed uniformly in a sphere of radius R , then

$$\rightarrow \text{the charge density} = \rho(r) = \frac{e}{\frac{4}{3}\pi R^3} \quad r < R$$
$$= 0 \quad r > R.$$

$$\rightarrow \text{the Electric field} = \vec{E} = \frac{(\text{charge enclosed})}{r^2} \hat{r}.$$

$$\vec{E}(r) = \frac{e r}{R^3} \hat{r} \quad r < R$$
$$= \frac{e}{r^2} \hat{r} \quad r > R.$$

$$\rightarrow \text{Potential energy} = V(r) = -(-e) \int_{\infty}^r \vec{E} \cdot d\vec{r}$$

$$V(r) = \int_r^{\infty} \frac{e^2}{r^2} dr \quad r > R$$
$$= \int_{\infty}^R \frac{e^2}{r^2} dr + \int_R^r \frac{e^2 r}{R^3} dr \quad r < R$$

$$V(r) = -\frac{e^2}{r} \quad r > R$$

$$= -\frac{e^2}{R} - \frac{e^2}{R^3} \left(\frac{R^2}{2} - \frac{r^2}{2} \right) \quad r < R$$

$$V(r) = \frac{3}{2} \frac{e^2}{R} + \frac{e^2 r^2}{2R^3} \quad r < R$$

$$-\frac{e^2}{r} \quad r > R.$$

We can write the Hamiltonian as

$$H = \underbrace{\frac{\vec{p}^2}{2\mu} - \frac{e^2}{r}}_{H_0} + V_{\text{pert}}(r)$$

where

$$V_{\text{pert}}(r) = \begin{cases} \frac{e^2 r^2}{2R^3} - \frac{3e^2}{2R} + \frac{e^2}{r} & r < R \\ 0 & r > R \end{cases}$$

The ground state (ignoring spin) is non degenerate.

The first order shift (E_0^1) of the ground state energy is

$$E_0^1 = \langle 100 | V_{\text{pert}} | 100 \rangle$$

$$\psi_{100}(r, \theta, \phi) = \frac{1}{(\pi a_0^3)^{1/2}} e^{-r/a_0}$$

$$= \int_0^R dr r^2 d\Omega \psi_{100}^*(r, \theta, \phi) \left(\frac{e^2 r^2}{2R^3} - \frac{3e^2}{2R} + \frac{e^2}{r} \right) \psi_{100}(r, \theta, \phi)$$

$$= \int_0^R dr r^2 \int d\Omega \left(\frac{e^2 r^2}{2R^3} - \frac{3e^2}{2R} + \frac{e^2}{r} \right) \frac{1}{(\pi a_0^3)} e^{-2r/a_0}$$

we can approx: $e^{-2r/a_0} \approx 1$ (Since $r < R$)

$$E_0^1 = \frac{4\pi}{\pi a_0^3} \cdot e^2 R^2 \cdot \left(\frac{1}{10} - \frac{3}{2} \cdot \frac{1}{3} + \frac{1}{2} \right)$$

$$E_0^1 = \frac{2}{5} \frac{e^2 R^2}{a_0^3}$$